Sahaj Bhadja

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Summary

Software engineer pursuing an MSCS at USC with 2+ years building **object-oriented backend systems**, **relational databases**, **automated tests**, and AI-enabled web tools across Java, Python, JavaScript, and cloud workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Designed and optimized relational data services for 500+ users, reducing request latency from **1.2s to 350ms** by tuning schemas, SQL access patterns, indexing strategies, and backend execution paths.
- Built automated validation and ETL checks for 5,000+ monthly jobs, reducing downstream failures and accelerating incident triage through earlier detection of regressions, data-quality drift, and runtime anomalies.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed React- and Flask-based features handling 10,000+ daily requests, translating product requirements into maintainable APIs and interfaces while improving delivery predictability through clearer ownership, testing, and sequencing.
- Built AI-assisted grading workflows with multimodal models, OCR, and relational persistence, reducing feedback turnaround by **60%** while supporting production-scale instructor operations in a fast-moving startup environment.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built Python experimentation pipelines across 100,000+ trajectory records, enabling reproducible algorithm evaluation and high-confidence benchmarking while improving signal quality for debugging model-driven decision workflows internally.
- Produced technical reports and targeted test scenarios translating findings into actionable guidance, improving cross-team alignment, reducing handoff ambiguity, and strengthening engineering quality as prototypes matured toward production readiness.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered object-oriented API and ETL microservices using Python, AWS Glue, and S3, improving response time by **30%** while supporting distributed integrations and stable delivery across production workloads.
- Maintained 20+ automated test suites using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and improving confidence in frequent releases across coordinated service components.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed MERN and WebSocket modules for 2M+ synchronized records across eight workflows, improving throughput consistency, reducing stale-session behavior, and strengthening real-time responsiveness for high-volume user-facing processes.
- Automated Docker-based build and test flows with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing environment drift that previously slowed validation, troubleshooting, and handoffs.

Skills

Languages: **Java**, **Python**, **C++**, JavaScript/TypeScript, SQL, Bash
Backend & Data: Spring Boot, Flask, FastAPI, REST APIs, PostgreSQL, MySQL, SQLite, relational schema design
Frontend: React, Next.js, Angular.js, responsive web interfaces, component-driven workflows
Cloud & Delivery: AWS, Docker, Kubernetes, Linux, Git/GitHub, Jenkins, CI/CD, automated testing

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating query authoring and schema exploration through a unified Java and TypeScript codebase with Docker-backed development workflows.

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*

- Implemented FCFS, SJF, SRT, RR, priority, and MLQ scheduling with **C++** logic, React controls, and Gantt visualization, improving algorithm debugging efficiency by **20%** across simulated workloads.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Built a **containerized REST processing API** on AWS CodePipeline and ECS/Fargate with automated validation gates, improving reliability, reducing deployment variance, and streamlining troubleshooting across production ETL workflows.